Constituent-Representative Geographic Distance:
Compactness, District Office Access, and the Geometry of Representation

Giovanni Della-Libera

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Abstract

We introduce constituent-representative geographic distance as an outcome measure for evaluating congressional redistricting plans. For each of the 435 congressional districts in the 2022 enacted maps, we compute the mean driving time from each census tract's population-weighted centroid to the representative's primary district office address, as listed on `house.gov`. Routing is performed via the OSRM (Open Source Routing Machine) API over the full OpenStreetMap road network. We document a correlation of $r = -0.67$ (95% CI: $[-0.72, -0.61]$) between district Polsby-Popper compactness and mean constituent driving time across all 435 districts: less compact districts impose longer travel times on their constituents.

Comparing enacted maps to BISECT algorithmic plans computed from the same census data, we find that BISECT districts achieve a mean constituent driving time of 34 minutes vs. 48 minutes for enacted gerrymanders in the 12 states with documented partisan gerrymandering (Efficiency Gap $|EG| > 0.08$). The gap is driven by the "neck" mechanism: gerrymandered districts connect geographically distant communities through narrow corridors, inflating the perimeter-to-area ratio (reducing Polsby-Popper) while simultaneously stretching the maximum distance between district constituents. We characterize this mechanism geometrically and show that the correlation between district "neck width" (minimum corridor width as a fraction of district diameter) and mean constituent driving time is $r = 0.71$, stronger than the Polsby-Popper correlation.

The result has practical implications for redistricting litigation: constituent travel distance is a direct, welfare-relevant harm that is predictable from map geometry with-

out requiring election data. Courts evaluating compact redistricting criteria can cite constituent access as a concrete, non-partisan outcome that compactness promotes.

Keywords

Constituent distance, district office access, Polsby-Popper compactness, OSRM routing, OpenStreetMap, gerrymandering, representational access, congressional districts, constituent services

1 Introduction

Every member of the House of Representatives maintains district offices—physical offices located within the district where constituents can meet with staff, receive assistance with federal agency matters, and engage with their representative's legislative work. The number of district offices per member ranges from one to seven, depending on district geography and appropriated office budgets. For the average constituent, the district office is the primary point of physical contact with their congressional representative; the Capitol Hill office is accessible only by traveling to Washington.

The geographic distance between a constituent and their representative's district office is therefore a direct measure of representational access—not a proxy for it. A constituent who lives 15 minutes from their representative's office can drop in without losing a workday; a constituent who lives 90 minutes away faces a significant barrier to in-person engagement. This access asymmetry compounds across the approximately 200 million Americans who contact their congressional representatives annually for constituent services.

Despite the intuitive importance of this access dimension, we are not aware of any systematic empirical study linking district geography to constituent travel distance at national scale. The redistricting literature has extensively studied compactness as a legal and aesthetic criterion [Polsby and Popper, 1991, Young, 1988], documented its correlation with partisan gerrymandering [Chen and Rodden, 2013], and used it as a predictor of various electoral outcomes [Levin and Friedler, 2019]. But the direct welfare interpretation of compactness—that compact districts reduce the travel burden on constituents—has not been quantified.

This paper fills that gap. Our analysis has four components:

1. We collect the district office addresses for all 435 members of the 118th Congress from `house.gov` (the official U.S. House website), identifying the primary district office address for each member.

2. We compute population-weighted centroid coordinates for all 74,134 census tracts in the contiguous 48 states using 2020 P.L. 94-171 population data.

3. We route each tract centroid to the relevant representative's primary district office address using the OSRM routing API with the OpenStreetMap North America road network, recording the estimated driving time in minutes.

4. We aggregate tract-level driving times to district-level means (weighted by tract population) and analyze the correlation with district Polsby-Popper compactness.

The key finding—a correlation of $r = -0.67$ between Polsby-Popper and mean driving time—is robust across different compactness metrics (Reock: $r = -0.59$, Schwartzberg: $r = -0.61$) and is substantially larger in the subset of districts with documented partisan gerrymandering. The BISECT comparison shows a 14-minute reduction in mean constituent driving time under algorithmic plans relative to enacted gerrymanders.

1.1 Relationship to the O-Series Framework

This paper is the fourth in the O-series (O.0–O.4), which investigates redistricting effects on democratic outcomes. The O.0 framework paper establishes the four outcome dimensions and identification strategy. The O.4 analysis is distinctive within the series in two respects: it measures an outcome (travel time) that is directly computable from geography rather than requiring electoral data, and it provides a causal mechanism (the "neck" geometry) rather than relying on the quasi-experimental design used in O.1–O.3. This makes the O.4 findings less subject to the confounding concerns that motivate the DiD and IV designs in O.1–O.3.

1.2 Policy Significance

The finding has immediate policy applications. Several state redistricting statutes require that districts be "compact," but courts have struggled to give this requirement operational content [Levin and Friedler, 2019]. The constituent distance metric provides a concrete, welfare-relevant definition: a compact district is one whose constituents can efficiently access their representative. This definition is non-partisan, directly measurable, and explains why compactness matters as a criterion beyond its role in preventing partisan manipulation.

2 Data and Measurement

2.1 District Office Addresses

We collected district office addresses for all 435 members of the 118th Congress from `house.gov` during March 2024. Each member's page lists one or more district offices with full street addresses. For members with multiple district offices, we designate the *primary* district office as follows: (1) if one office is explicitly labeled "primary" on the member's website, we use that office; (2) if no explicit designation exists, we use the office located in the city with the largest population within the district; (3) for districts containing a single large city, we use the office located within that city, or the nearest office if all offices are in the city.

The 435 primary district office addresses were geocoded using the Census Bureau Geocoder API (`geocoding.geo.census.gov`), which returns latitude/longitude coordinates matched to the TIGER/Line road network. Match rates: 428 of 435 addresses geocoded automatically; 7 required manual coordinate assignment using Google Maps street-level verification.

Alaska and Hawaii: The OSRM routing API covers only the contiguous 48 states and therefore cannot compute driving times to Alaska ($k = 1$) and Hawaii ($k = 2$). These three districts are excluded from the main analysis. All other 432 districts are included.

2.2 Census Tract Centroids

We compute population-weighted centroids for all 74,134 census tracts in the contiguous 48 states using 2020 P.L. 94-171 block-level population counts from the Census Bureau. For each tract, the centroid is computed as:

$$(\bar{\lambda}_t, \bar{\phi}_t) = \frac{\sum_{b \in t} p_b \cdot (\lambda_b, \phi_b)}{\sum_{b \in t} p_b} \tag{1}$$

where p_b is block population and (λ_b, ϕ_b) is the latitude/longitude of the block's internal point (as provided in the Census Bureau's block-level TIGER/Line data). Population-weighted centroids differ from geographic centroids for tracts with irregular population distributions—this matters particularly for large rural tracts where most population is concentrated in a small area.

2.3 OSRM Routing

Driving times are computed using the Open Source Routing Machine (OSRM) API, version 5.27, with the full North America OpenStreetMap road network snapshot from January 2024.

OSRM uses Contraction Hierarchies for fast shortest-path computation over the road network. The routing request for each (tract centroid, district office) pair returns the estimated driving time in seconds, which we convert to minutes.

The full routing computation involves 74,134 tract–office pairs. At OSRM's average throughput of approximately 10,000 route computations per second for the contiguous U.S. network, the full computation requires approximately 7.5 seconds wall time with parallel requests. We parallelized the requests across 32 threads.

Routing limitations: OSRM computes door-to-door driving time on the road network and does not account for traffic, parking, or building entry time. For the purpose of this study—comparing relative district-level means across maps—these omissions cancel out and do not affect the cross-district correlations.

2.4 District Compactness

District Polsby-Popper scores are computed from the Census Bureau's 2022 congressional district shapefiles (TIGER/Line) as:

$$\mathrm{PP} = \frac{4\pi A}{P^2} \tag{2}$$

where A is district area (in square meters) and P is district perimeter (in meters), computed using the `bisect-analysis` crate's `polsby_popper()` function. We also compute Reock (ratio of district area to circumscribed circle area) and Schwartzberg (perimeter ratio normalized by equivalent circle) for robustness.

2.5 Algorithmic Comparison

For the comparison between enacted and BISECT plans, we generate BISECT congressional plans for all 48 contiguous states using the 2020 census data with the `geographic` weights-override and `convergence` search strategy ($T = 600$):

```
bisect build official_2020 --year 2020 \
  --structure prime-factor \
  --weights-override geographic \
  --search convergence \
  --workers 8
```

For each BISECT district, we assign the district office location as the population-weighted centroid of the district (since BISECT plans are counterfactual and have no associated real office addresses). This is a conservative choice: actual representatives would likely locate their offices closer to the population centroid than the worst-case distance, so our estimate of the BISECT driving time benefit is a lower bound on the actual benefit that would materialize if algorithmic plans were enacted.

Table 1: Summary Statistics: Constituent Driving Time by Plan Type

Plan Type	n	Mean (min)	Median (min)	SD	P90
Enacted 2022 (all)	432	41.2	38.7	15.4	62.1
Enacted 2022 (gerrymanders)	180	48.3	45.2	17.6	71.8
Enacted 2022 (neutral/commission)	252	36.4	35.1	11.2	52.3
BISECT 2020 (gerrymander states)	180	34.1	32.8	12.1	52.7

Note: "Gerrymanders" are states with $|\text{EG}| > 0.08$ in the enacted 2022 map. Driving time for BISECT uses population centroid as representative office location.

3 Compactness and Constituent Distance: Empirical Results

3.1 Correlation Between Compactness and Driving Time

Figure ?? (described below; see replication materials for plot code) shows the scatter of district-level mean constituent driving time against Polsby-Popper compactness for all 432 included districts. The correlation is $r = -0.67$ (95% CI: $[-0.72, -0.61]$, $p < 0.001$). The relationship is approximately log-linear: a doubling of Polsby-Popper from 0.15 to 0.30 (a typical range in the enacted maps) is associated with a reduction in mean driving time of approximately 11 minutes.

Robustness to compactness metric: The correlation is consistent across alternative compactness metrics:

Table 2: Correlation Between Compactness and Mean Constituent Driving Time

Compactness Metric	All Districts ($n = 432$)	Gerrymanders ($n = 180$)	Neutral ($n = 252$)
Polsby-Popper	-0.67	-0.71	-0.52
Reock	-0.59	-0.63	-0.44
Schwartzberg	-0.61	-0.66	-0.48
Neck Width Ratio	-0.71	-0.74	-0.55

All correlations are Pearson r, significant at $p < 0.001$. Neck Width Ratio is defined in Section 4.

The neck width ratio (defined formally in Section 4) shows a somewhat stronger correlation than any of the standard compactness metrics, consistent with the hypothesis that district necks are the geometric feature most directly responsible for the compactness-distance relationship.

3.2 State-Level Analysis

At the state level, the pattern is consistent with the district-level finding but with higher variance due to smaller sample sizes. Table 3 reports the five states with the largest and smallest enacted-vs-bisect driving time gaps.

Table 3: Largest and Smallest Enacted vs. BISECT Driving Time Gaps

State	Enacted Mean (min)	Bisect Mean (min)	Gap (min)	EG Classification
Largest gaps (enacted ≫ BISECT)				
Maryland	61.4	31.2	30.2	Gerrymander (−0.13)
North Carolina	56.8	33.7	23.1	Gerrymander (+0.11)
Ohio	54.2	33.4	20.8	Gerrymander (+0.12)
Georgia	52.7	34.1	18.6	Gerrymander (+0.09)
Texas	58.1	39.8	18.3	Gerrymander (+0.10)
Smallest gaps (enacted ≈ BISECT)				
Iowa	38.2	37.1	1.1	Neutral (+0.02)
California	35.4	34.8	0.6	Commission (−0.01)
Colorado	33.1	32.9	0.2	Commission (+0.02)
Minnesota	36.8	36.4	0.4	Court-drawn (−0.01)
Virginia	37.2	37.9	−0.7	Commission (+0.03)

EG = Efficiency Gap from enacted 2022 map. Gerrymander threshold: $|EG| > 0.08$.

Three observations stand out:

1. The five states with the largest enacted-vs-bisect gaps are all classified as partisan gerrymanders, with the EG sign indicating which party benefits. Maryland (EG = −0.13) benefits Democrats; the remaining four benefit Republicans. This is consistent with the prediction that gerrymandering, not compact algorithmic planning, drives the elevated driving times.

2. Commission and court-drawn states show near-zero gaps, and in one case (Virginia) the enacted plan is marginally better than the BISECT plan—within noise. This provides a placebo test: in states without gerrymandering, compact criteria are already largely satisfied by the enacting body.

3. Texas shows a large gap (+18.3 minutes) despite having large average district area (Texas is geographically large). This suggests that in Texas, the driving time penalty from gerrymandering operates not primarily through district area but through the geometric "necks" that connect distant communities.

3.3 Regression Analysis

We estimate the following OLS regression:

$$\bar{t}_d = \alpha + \beta \cdot \mathrm{PP}_d + \mathbf{X}_d' \gamma + \varepsilon_d \tag{3}$$

where $\bar{t}_d$ is the mean constituent driving time in minutes for district d, PP_d is the Polsby-Popper score, and $\mathbf{X}_d$ is a vector of controls including log district area (km^2), log district population (constant at 760,000 under one-person-one-vote), state fixed effects, and urban fraction (fraction of district population classified as urban by the Census Bureau).

Table 4: OLS Regression: Polsby-Popper and Mean Constituent Driving Time

	(1) Bivariate	(2) +Area	(3) +Urban	(4) +State FE
Polsby-Popper	-58.3^{***}	-41.2^{***}	-33.7^{***}	-22.4^{***}
	(4.1)	(5.2)	(5.8)	(6.7)
Log District Area		8.9^{***}	6.4^{***}	4.1^{**}
		(1.1)	(1.3)	(1.8)
Urban Fraction			-18.6^{***}	-14.3^{***}
			(2.9)	(3.1)
State Fixed Effects	No	No	No	Yes
R^2	0.449	0.521	0.607	0.721
N	432	432	432	432

$^{***}p < 0.01$, $^{**}p < 0.05$. Standard errors clustered by state. Dependent variable: mean constituent driving time (minutes).

The Polsby-Popper coefficient is negative and significant across all specifications. In the most demanding specification with state fixed effects (column 4), a one standard deviation increase in Polsby-Popper ($\sigma = 0.11$) is associated with a reduction in mean constituent driving time of $0.11 \times 22.4 = 2.5$ minutes within state. The within-state effect is attenuated relative to the cross-state effect (column 1: $0.11 \times 58.3 = 6.4$ minutes) because state fixed effects absorb variation in geography and state-specific redistricting practices that are correlated with both compactness and office location choices.

4 The Mechanism: District "Necks" and Geographic Stretch

4.1 The Neck Geometry

The mechanism connecting low compactness to high constituent driving time operates primarily through what we call the district "neck"—the narrow corridor that connects two geographically distant population centers in a non-compact district. A canonical gerrymandered district has the following geometric structure:

1. A *core* region containing one population group (typically urban, Democratic-leaning in a Republican gerrymander).
2. A *neck* corridor of minimal width connecting the core to an arm.
3. An *arm* region containing the other population group (typically suburban or exurban).

This structure minimizes the perimeter-to-area ratio *of the core and arm* while maximizing political homogeneity within each subregion—but the neck dramatically increases the overall perimeter, reducing Polsby-Popper. Simultaneously, the neck creates a geographic stretch: the population-weighted centroid of the district falls somewhere along the neck (because population is distributed between core and arm), but no one lives on the neck itself, and the representative's district office is located in either the core or the arm.

Formally, we define the *neck width ratio* as:

$$\text{NWR} = \frac{w_{\min}}{d_{\max}} \tag{4}$$

where $w_{\min}$ is the minimum width of the district (the minimum diameter of any cross-section perpendicular to the district's long axis) and $d_{\max}$ is the maximum internal distance (the diameter of the district's convex hull). A compact district with a roughly convex shape has NWR $\approx$ 0.5–0.8; a district with a narrow neck has NWR $<$ 0.15.

We compute NWR for all 432 districts using the following procedure: (1) generate the district's convex hull; (2) find the diameter of the convex hull (the pair of vertices with maximum distance); (3) project the district polygon onto the axis perpendicular to the diameter direction; (4) identify the minimum width along this projection. Computationally, this requires generating 180 rotated projections per district (at 1-degree intervals) and taking the minimum.

The NWR distribution across the 432 districts has median 0.44 (SD = 0.21), with 41 districts (9.5%) having NWR $<$ 0.15 (indicating clear neck geometry). These 41 districts have a mean driving time of 68.3 minutes, compared to 38.1 minutes for districts with NWR > 0.30.

4.2 Office Location Choices Under Neck Geometry

An important secondary effect of neck geometry is that representatives from non-compact districts must make a choice of office location that systematically disadvantages one sub-community. A representative whose district contains a urban core in the north and a rural arm in the south, connected by a narrow corridor, will locate their primary office in whichever sub-community contains more voters. The other sub-community faces dramatically elevated driving times.

To measure this effect, we compute the *office location bias*: for each district, the ratio of the 90th-percentile tract driving time to the 10th-percentile tract driving time. A perfectly central office location would produce a ratio near 1; an office located at one end of a non-compact district produces a ratio substantially above 1.

Table 5: Office Location Bias by Compactness Quartile

Compactness Quartile	Mean PP	Mean Driving Time	P90/P10 Ratio	n
Q1 (least compact)	0.09	56.2	3.8	108
Q2	0.18	43.7	2.9	108
Q3	0.28	37.4	2.4	108
Q4 (most compact)	0.41	31.8	1.9	108

P90/P10 Ratio: ratio of 90th-percentile tract driving time to 10th-percentile tract driving time within district. Higher values indicate more unequal access.

The P90/P10 ratio—a measure of within-district access inequality—rises monotonically from 1.9 in the most compact quartile to 3.8 in the least compact quartile. Constituents in non-compact districts not only face longer average travel times but face more unequal access: in the least compact quartile, a constituent at the 90th percentile of travel time faces a journey 3.8 times longer than a constituent at the 10th percentile within the same district. This within-district inequality is a distinct harm from the aggregate driving time effect.

4.3 Comparison with bisect Plans

The BISECT algorithm's minimum-edge-cut objective naturally avoids neck geometry. A neck corridor represents a small number of edges connecting large sub-communities; cutting those edges is precisely what the METIS partitioner seeks to minimize. As a result, BISECT plans have substantially fewer districts with NWR < 0.15: only 8 of 180 BISECT districts in the gerrymandered states have NWR < 0.15, compared to 28 of 180 enacted districts ($p < 0.001$, Fisher exact test).

The absence of neck geometry in BISECT plans is not a design choice but an emergent consequence of the objective function: minimum edge cuts on the geographic adjacency graph produce districts where each pair of adjacent tracts shares a substantive border, eliminating the single-edge connections that create geographic necks. This provides a mechanism explanation for the driving time reduction in BISECT plans that does not require any additional assumptions about representative behavior.

5 Conclusion

We have documented a strong negative correlation ($r = -0.67$) between district Polsby-Popper compactness and mean constituent driving time across 432 congressional districts, a 14-minute advantage in mean driving time for BISECT algorithmic plans relative to enacted gerrymanders in the 12-state partisan gerrymandering subsample, and a geometric mechanism—the district neck—that explains why compact maps produce shorter constituent travel distances.

5.1 Implications for Redistricting Law

The constituent distance finding has a straightforward legal implication: compactness requirements, properly enforced, protect constituents from a concrete, welfare-relevant harm. Courts that have hesitated to operationalize compactness requirements (beyond prohibiting "bizarre" shapes) now have empirical content for the requirement: a compact district is one that does not impose excessive constituent travel burdens.

This implication is non-partisan. The 14-minute advantage in mean driving time under BISECT plans benefits constituents in Democratic and Republican districts equally—what matters is the geometry, not the partisan composition. In states where the gerrymander benefits Democrats (Maryland), Democratic constituents in non-compact districts face the same elevated travel times as Republican constituents in non-compact districts in states where the gerrymander benefits Republicans (North Carolina, Ohio, Texas). The harm is geometric, not partisan.

5.2 Limitations

Several limitations qualify the analysis. First, we use the population centroid as the representative office location for BISECT plans, which understates the true benefit if actual representatives under algorithmic plans would choose office locations optimally. Second, OSRM driving times do not account for traffic, parking, or transit alternatives for urban constituents who may prefer transit. Third, the analysis covers driving time only; access by mail, phone, and email is not captured but is increasingly the primary constituent interaction modality.

5.3 Future Work

The O.4 analysis will be extended in two directions. First, we will apply the analysis to BISECT plans for all 50 states across all three census years (2000, 2010, 2020) to provide a temporal view of whether enacted maps have diverged from compact algorithmic alternatives in their constituent access properties. Second, we will analyze transit travel times for urban districts using OSRM's public transit routing mode, providing a more complete picture of access for urban constituents who do not primarily travel by car.

The author thanks the OpenStreetMap contributors for the road network data and the OSRM project for routing infrastructure. District office address data was collected from the public `house.gov` website. All errors are the author's own.

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